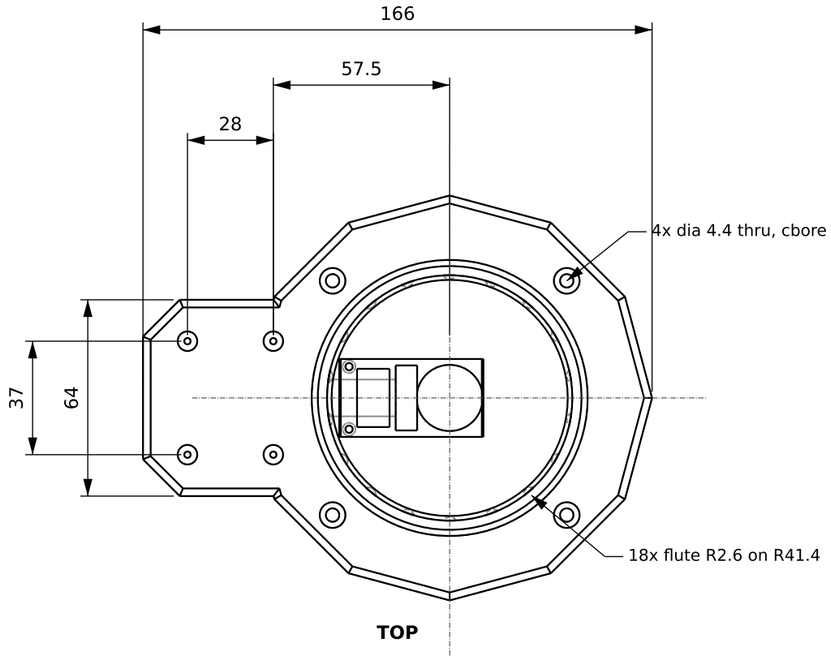
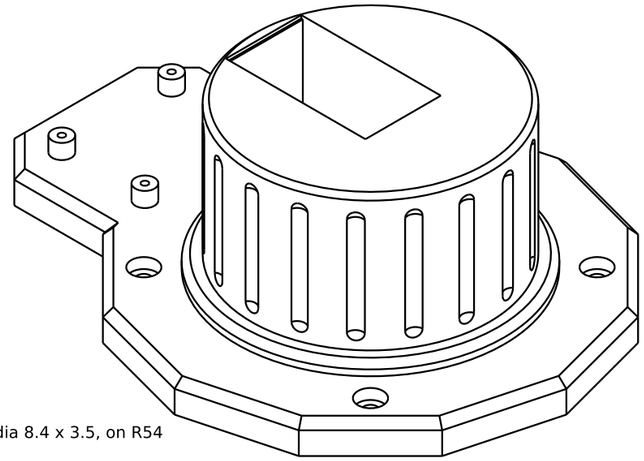
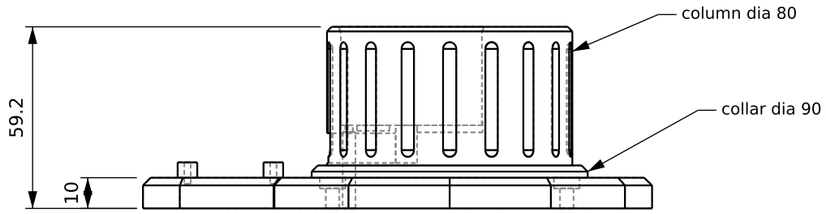


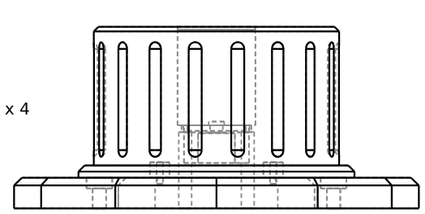
TOP



ISOMETRIC (NOT TO SCALE)



FRONT



RIGHT

FEATURES (in order)

1. Plinth: regular 12-gon, circumscribed dia 132 (a corner on +X), extrude z 0..10.
2. Board deck: polygon (-40,-32) (-88,-32) (-100,-20) (-100,20) (-88,32) (-40,32), extrude z 0..10; union with the plinth.
3. Chamfer 2.5 on all top outer edges (z = 10).
4. Mounting holes: 4x dia 4.4 through on R54 at 45/135/225/315 deg; counterbore dia 8.4 from z 6.5 up.
5. Collar: cylinder dia 90, z 10..14; chamfer 2 on its top edge.
6. Column: cylinder dia 80, z 10..59.2; chamfer 1.5 on its top edge.
7. Board standoffs: 4x cylinder dia 6.4, z 9..15, at (-57.5,+/-18.5) and (-85.5,+/-18.5); pilot dia 2.1 from z 8 to 16 (M2.5 self-tapping).
8. Flutes: 18x vertical cylinder dia 5.2 on R41.4 at 10 deg + 20 deg steps, z 19..52, with dia 5.2 spheres at both ends; cut.
9. Servo pocket P (sheet TAL-001) with the servo frame origin at (0, 0, 43.3); servo +Z (shaft) -> +Z, servo +X -> +X, servo +Y -> +Y.
10. Top opening: cut box x -36..11, y -12.7..12.7, z 58.2..64.2.
11. Cable tunnel (plug bay -> board deck): cut box x -60..-13, y -6..6, z 14.8..24.5.
12. Case screw holes (sheet TAL-002): connector face only, screws from below: (x, y) = (-32.8, +/-10.2); hole z 24.9..27.5, counterbore z -1.7..24.9 (opens on the bottom). Each is a dia 2.4 hole from the servo face to the seat, then a dia 4.2 counterbore from the seat outward through the part.

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- The pan servo drops in from the top; its plugs hang into the bay under it.

WORK SET A: base, shoulder, upper arm, hub caps

THIRD ANGLE

PROJECT TALOS 5-axis arm	EST. MASS 155 g
PART **Base**	SCALE 1:2
PART NO. **TAL-101**	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-03
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 4 of 12